A Completion-Aware Framework for Impactful Counterfactual Explainability in Graph Neural Networks

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A Appendix

The Appendix first presents the datasets and their statistics (Section A.1) and then defines the evaluation metrics together with their mathematical formulations (Section A.2). Furthermore, it describes the training configurations of the oracle and reconstruction models (Sections A.3 and A.4) and reports the hyperparameter settings of our method, along with the rationale behind them and additional notes on the denoising and robustness experiments (Section A.5). Detailed experimental results, including per-class metrics and further insight into the OH module, are provided in Section A.6, while Section A.7 reports additional experiments on the MUTAG dataset. Finally, the Appendix summarizes the experimental configurations of the competing baseline methods (Section A.8) and describes the hardware used to run the experiments (Section A.9).

A.1 Datasets

Table 1 summarizes the statistics of the datasets used in our experiments.

Table 1: Dataset statistics.

Dataset	# of classes	# of features	Avg. # of nodes	Avg. # of edges	# of train graphs	# of val. graphs	# of test graphs
BA-2Motifs	2	10	25	25,49	800	100	100
BA-2Motifs-3Classes	3	10	25	25,23	800	100	100
BA-3Motifs	3	10	25	27,06	800	100	100
BA-4Motifs	4	10	25	27,09	800	100	100
BBBP	2	9	24.06	25,35	1631	203	205
Mutag	2	10	30.26	30,92	1840	230	231
Twitter	3	768	21.103	20,35	4,998	1,250	692
Graph-SST5	5	768	19.849	18,66	8,544	1,101	2,210

The **BA-2Motifs** [3] dataset is a synthetic graph classification dataset. Each graph is created by attaching either a five-node cycle motif or a five-node house motif to a base graph generated using the Barabási–Albert (BA) model. Graphs are labeled 0 (cycle) or 1 (house), and all node features are 10-dimensional vectors with constant value 0.1.

Building on this dataset, we construct **BA-3Motifs** and **BA-4Motifs** datasets by modifying the original five-node motif subgraph in a subset of graphs. In BA-3Motifs, we introduce an additional motif type-a five-node clique-creating a third class with label 2. In BA-4Motifs, we further extend the dataset to four motif types: the original five-node cycle (label 0) and house motif (label 1), together with a four-node clique (label 2) and a five-node clique (label 3).

For **BA-2Motifs-3classes**, we follow the same graph generation process as BA-2Motifs and randomly remove a single edge from the motif in one third of the graphs. We assign a third label (label 2) to this class of partially complete graphs.

Graph-SST5 and **Graph-Twitter** [6] are graph classification datasets, with 5, and 3 classes, respectively. They are designed for sentiment analysis tasks, where the highest class corresponds to the most positive sentiment. Each graph represents a sentence, nodes represent words, and edges represent relationships between words. Node features, extracted using a pre-trained BERT model.

BBBP [4] is a molecular graph classification dataset, used to predict the permeability of Blood-Brain Barrier (BBB). Each compound is represented as a graph, where atoms are represented as nodes and bonds as edges. Each node has a size vector 9 derived from the molecular structure of the compound, and each graph is labeled 0 or 1 to indicate the permeability of the blood-brain barrier.

A.2 Evaluation Metrics

Validity : Validity (sometimes referred to as accuracy or correctness) measures the fraction of graphs for which there exists at least one counterfactual graph, i.e., a graph whose oracle prediction differs from that of the original graph. Formally,

$$\text{Validity} = \frac{1}{M} \sum_{G_i \in \mathcal{G}} \begin{cases} 1, & \text{if } \exists G_i^{cf,j} \text{ , s.t. } \Phi(G_i^{cf,j}) \neq \Phi(G_i), \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $G_i^{cf,j}$ denotes the j-th counterfactual graph generated for the input graph G_i, and $M = |\mathcal{G}|$ denotes the number of input graphs.

Explanation Size : The explanation size is the total number of edge removals and additions that constitute the counterfactual explanation. It is defined as:

$$ExpSize(G_i, G_i^{cf,j}) = \sum_{u,v} |a_{u,v} - a_{u,v}^{cf}|, \tag{2}$$

where $G_i = \{\mathbf{A}_i, \mathbf{X}_i\}$ denotes the input graph and $G_i^{cf,j} = \{\mathbf{A}_i^{CF,j}, \mathbf{X}_i\}$ denotes the counterfactual graph j (in case multiple counterfactual explanations are generated for the same input graph). Moreover, $a_{u,v}$ and $a_{u,v}^{cf}$ denote the entries of the adjacency matrices $\mathbf{A}_i$ and $\mathbf{A}_i^{CF,j}$, respectively, corresponding to nodes u and v.

Fidelity : Fidelity measures the decrease in the confidence of the oracle after applying counterfactual modifications. It is defined as:

$$Fidelity(G_i, G_i^{cf,j}) = \Phi(G_i) - \Phi(G_i^{cf,j}), \tag{3}$$

where $\Phi(\cdot)$ denotes the confidence of the oracle for the original prediction class.

Motif Proximity : A quality criterion for a CFX to be intuitive for humans is to be relevant to the input, which in the case of graphs can be seen as the ability to modify edges that adhere to the motif that generates the original prediction. Motif Proximity calculates the proportion of edges in the explanation that are connected to the motif of the input graph. Let G_i denote the input graph and $G_i^{cf,j}$ denote the $j-th$ counterfactual graph, in case multiple counterfactual explanations are generated. Let also $\mathcal{E}_{i,j}^{diff}$ denote the set of edge modifications (added or removed) applied to G_i that lead to $G_i^{cf,j}$. Thus, $\mathcal{E}_{i,j}^{diff}$ denotes the counterfactual explanation. The Motif Proximity is defined as:

$$\text{MotifProx} = \frac{\sum_{e \in \mathcal{E}_{i,j}^{diff}} \begin{cases} 1, & \text{if } e \text{ touches at least one motif node,} \\ 0, & \text{otherwise,} \end{cases}}{ExpSize(G_i, G_i^{cf,j})}. \tag{4}$$

Minimality : Minimality evaluates whether the edits of an explanation are necessary to change the oracle's prediction. This metric measures the proportion of subsets of edits that do not change the original prediction. Higher values indicate that the edits in the explanation are necessary, thus producing more minimal counterfactual explanations. Intuitively, edits are considered necessary when their removal prevents the explanation from producing a counterfactual. Let $\mathcal{E}_{i,j}^{diff}$ denote the set of edge modifications (added or removed) applied to G_i that lead to $G_i^{cf,j}$. Thus, $\mathcal{E}_{i,j}^{diff}$ denotes the counterfactual explanation. Let $\mathcal{P}(\mathcal{E}_{i,j}^{diff})$ denote the set of all non-empty subsets of these edits. For a subset $S \in \mathcal{P}(\mathcal{E}_{i,j}^{diff})$, let G_i^S be the graph obtained by applying only the edits in S to G_i. The Minimality score is defined as:

$$\text{Minimality} = \frac{1}{|\mathcal{P}(\mathcal{E}_{i,j}^{diff})|} \sum_{S \in \mathcal{P}(\mathcal{E}_{i,j}^{diff})} \begin{cases} 1, & \text{if } \Phi(G_i^S) = \Phi(G_i), \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Validity after Noise : Robustness of CfXs, i.e., the validity of explanations under changing conditions, is a major topic in the broader field of AI, but is rarely discussed in the context of counterfactual GNN explainability. According to [2], an important aspect of robustness is related to noise in the input, called Validity after Noise (VaN). Let $\mathcal{G}^{CF,i}$ and $\mathcal{G}_\sigma^{CF,i}$ be the sets of counterfactual graphs generated for the input graph G_i before and after the injection of noise, respectively.

We define x as the number of input graphs for which there exists at least one counterfactual in $\mathcal{G}_{\sigma}^{CF,i}$ with the same prediction change as a counterfactual in $\mathcal{G}^{CF,i}$, and n as the number of input graphs for which at least one counterfactual was generated before noise injection (which corresponds to the Validity score). Thus, VaN is defined as:

$$\mathrm{VaN} = \frac{x}{n}. \tag{6}$$

To quantify the uncertainty of this proportion, we report the Wilson confidence interval:

$$\frac{\mathrm{VaN} + \frac{z^2}{2n} \pm z\sqrt{\frac{\mathrm{VaN}(1-\mathrm{VaN})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{7}$$

where $z = 1.96$ corresponds to 95% confidence level.

Edge Consistency after Noise (ECaN) : Inspired by prior robustness evaluation protocols that compare explanations before and after perturbing the input graph [1], ECaN quantifies the structural stability of counterfactual explanations using the Jaccard similarity between the edge-edit sets obtained from the original and noisy inputs. Let $\mathcal{E}_{i,j}^{diff}$ denote the set of edge modifications (i.e., the counterfactual explanation) applied to the input graph G_i that lead to the counterfactual graph $G_i^{CF,j}$, where j indexes one of the generated counterfactuals in cases where multiple counterfactual explanations exist for the same input graph, G_i. Similarly, let $\mathcal{E}_{\sigma,i,k}^{diff}$ denote the set of edge modifications applied to the noisy graph G_i^{σ} that lead to the counterfactual graph $G_{\sigma,i}^{CF,k}$, where k indexes one of the generated counterfactual graphs for G_i^{σ}. If multiple counterfactual explanations are generated for either the original or noisy graph, we compute the Jaccard similarity for all pairs of explanations and retain the maximum value. Then, ECaN is defined as:

$$\mathrm{ECaN} = \frac{1}{x}\sum_{i=1}^{x} \max_{j,k} \frac{|\mathcal{E}_{i,j}^{diff} \cap \mathcal{E}_{\sigma,i,k}^{diff}|}{|\mathcal{E}_{i,j}^{diff} \cup \mathcal{E}_{\sigma,i,k}^{diff}|}. \tag{8}$$

A.3 Training the oracle

We report the training configuration of the oracle classifier used throughout our experiments. For all datasets, we employ a 3-layer GCN with ReLU activations. Node embeddings are aggregated via a readout function (mean or max), producing graph-level representations. Table 2 summarizes the dataset-specific hyperparameters and the corresponding test accuracy. The oracle architecture follows the design of [6].

A.4 Training the reconstruction model

Table 3 summarizes the hyperparameters used to train the reconstruction model. For all datasets, we employ the same base training configuration, using a batch

Table 2: Dataset-specific hyperparameters and achieved accuracy.

Dataset	Batch Size	Epochs	Readout	Size of layers	LR	WD	Accuracy
BA-2Motifs	64	800	mean	20, 20, 20	0.001	0.0	0.98
BA-2Motifs-3Classes	64	800	mean	20, 20, 20	0.001	0.0	0.82
BA-3Motifs	64	800	mean	20, 20, 20	0.001	0.0	0.94
BA-4Motifs	64	800	mean	20, 20, 20	0.001	0.0	0.94
BBBP	32	200	max	128, 128, 128	0.001	5e-4	0.8195
Mutag	32	200	max	128, 128, 128	0.001	5e-4	0.9870
Twitter	128	50	max	128, 128, 128	0.001	0.0	0.7066
Graph-SST5	128	50	max	128, 128, 128	0.001	0.0	0.5050

size of 16, 150 training epochs, and a projection dimension of $d' = 200$. Recall that the projection dimension refers to the dimensionality of the projected one-hot class representation (when One-Hot is enabled). The same hyperparameters are used for both One-Hot settings ($-$OH /+OH).

Table 3: Reconstruction model hyperparameters and validation AUC (ValAUC) for each dataset and One-Hot (OH) setting.

Dataset	OH	EncLay	EncDrop	EncHid	MLPLay	MLPdim	Disj	Neg	ValAUC
BA-2Motifs	-	4	0.05	200	4	4000/4000/2000	0.30	2.5	0.9295
	+								0.9334
BA-2Motifs-3Classes	-	4	0.05	200	4	4000/4000/2000	0.30	2.5	0.9165
	+								0.9191
BA-3Motifs	-	4	0.05	200	4	4000/4000/2000	0.30	2.5	0.9367
	+								0.9427
BA-4Motifs	-	4	0.05	200	4	4000/4000/2000	0.30	2.5	0.9444
	+								0.9437
BBBP	-	3	0.10	100	3	512/256	0.35	1.0	0.9449
	+								0.9472
Mutag	-	3	0.10	100	3	512/256	0.35	1.0	0.9661
	+								0.9667
Twitter	-	2	0.40	100	3	256/128	0.35	1.0	0.9330
	+								0.9350
Graph-SST5	-	2	0.40	100	3	256/128	0.35	1.0	0.9492
	+								0.9523

A.5 Hyperparameter Settings

Table 4 reports the dataset-specific hyperparameter settings of DR-CFGNN .

In all synthetic datasets, N' was set to 6, since the motifs consist of 5 nodes and we include an additional node corresponding to the connection node to the BA graph. The only exception is the BA-2Motifs-3class dataset, where we set it slightly higher, as this dataset is more challenging due to the random removal of motif edges from some graphs. We increase this parameter to make the extracted subgraph more robust. In the real-world datasets, N' is set to half of the average number of nodes per graph.

In synthetic datasets, we use a stricter threshold τ than in real-world datasets, since ground-truth explanations are available and allow for a more direct evaluation of the results; moreover, these datasets are comparatively simpler, as they primarily contain structural patterns and lack semantic information.

Here, R denotes the maximum number of edge removals considered during the deletion step, while K denotes the maximum number of edge additions considered during the addition step.

Table 4: Dataset-specific settings for DR-CFGNN .

Dataset	N'	α_{del}	β_{del}	R	α_{add}	β_{add}	K	τ
BA-2Motifs	6	0.5	1	2	0.5	1	2	0.9
BA-2Motifs-3Classes	8	0.5	1	2	0.5	1	2	0.8
BA-3Motifs	6	0.5	1	2	0.01	3	5	0.9
BA-4Motifs	6	0.5	1	2	0.01	3	5	0.9
BBBP	12	0.01	2	5	0.01	2	5	0.6
Mutag	20	0.1	0.5	1	–	–	–	–
Twitter	11	0.001	3	7	0.001	3	7	0.6
Graph-SST5	10	0.001	3	7	0.001	3	7	0.6

In the following, we state the rationale for the chosen values per dataset.

- **BA-2Motifs & BA-2Motifs-3Classes:** We set $R = 2$ and $K = 2$, allowing the deletion and addition of up to two edges. This choice reflects the fact that, in these synthetic datasets, transforming one motif pattern into another typically requires a single edge edit, and at most two in more challenging cases, such as transforming an incomplete cycle into the house motif, which requires adding up to two edges.
- **BA-3Motifs & BA-4Motifs:** We set $K = 5$, since transforming from the sparsest to the densest motif requires at most five added edges. We keep $R = 2$; this asymmetric choice prioritizes additive edits and limits deletions, which is aligned with our goal of generating compact counterfactuals. As a result, clique-to-sparser transformations are harder under this setting. If needed, such transitions can be supported by increasing R accordingly.
- **Twitter & Graph-SST5:** For these datasets, we increase both K and R to 7, with the center of the addition and deletion distributions set to 3, allowing a wider range of structural edits. Unlike the synthetic datasets, these graphs do not contain explicit motif-based patterns; instead, they involve more complex and less structured relationships. Therefore, a larger number of edge modifications are required to meaningfully alter the model's prediction.
- **BBBP:** We set $R = 5$ and $K = 5$ to allow a broader range of edits, since counterfactuals in molecular graphs often require multiple bond deletions and additions.
- **Mutag:** See Appendix A.7.

Table 5: Detailed counterfactual metrics

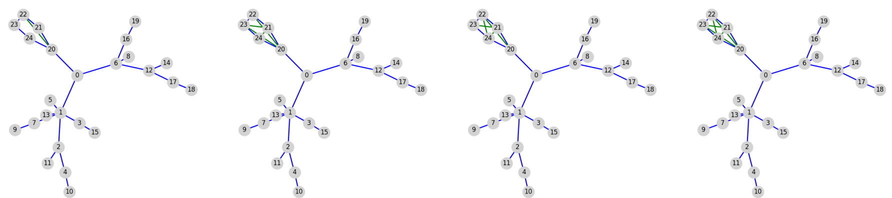
		-DeN-OH					-DeN+OH					+DeN-OH					+DeN+OH				
Dataset	Target	Validity	Fid	Size	Minim	Prox	Validity	Fid	Size	Minim	Prox	Validity	Fid	Size	Minim	Prox	Validity	Fid	Size	Minim	Prox
BA-2Motifs	class 0	-	-	-	-	-	0.860	0.96	1.45	1.00	0.80	-	-	-	-	-	0.877	0.97	1.40	1.00	0.85
	class 1	-	-	-	-	-	1.000	0.95	1.22	1.00	0.96	-	-	-	-	-	1.000	0.95	1.24	1.00	0.96
	Any	0.867	0.94	1.45	0.99	0.90	0.918	0.95	1.34	1.00	0.88	0.898	0.94	1.47	0.99	0.90	0.929	0.96	1.33	1.00	0.90
BA-2Motifs-3Classes	class 0	-	-	-	-	-	0.946	0.82	1.34	0.94	0.82	-	-	-	-	-	0.982	0.80	1.36	0.92	0.78
	class 1	-	-	-	-	-	0.959	0.74	1.34	0.80	0.81	-	-	-	-	-	0.980	0.74	1.38	0.80	0.81
	class 2	-	-	-	-	-	1.000	0.83	1.19	0.97	0.90	-	-	-	-	-	0.949	0.83	1.18	0.97	0.83
	Any	1.000	0.83	1.15	1.00	0.90	1.000	0.82	1.15	0.98	0.88	0.988	0.82	1.15	1.00	0.86	0.988	0.81	1.15	0.98	0.89
BA-3Motifs	class 0	-	-	-	-	-	0.359	0.77	1.17	1.00	1.00	-	-	-	-	-	0.312	0.78	1.15	1.00	1.00
	class 1	-	-	-	-	-	0.938	0.90	1.90	1.00	0.96	-	-	-	-	-	0.922	0.90	1.90	1.00	0.97
	class 2	-	-	-	-	-	0.967	0.84	3.45	0.52	0.97	-	-	-	-	-	0.900	0.85	3.44	0.48	0.97
	Any	0.904	0.85	1.72	1.00	0.93	0.936	0.86	1.77	1.00	0.97	0.894	0.86	1.83	1.00	0.94	0.883	0.87	1.77	1.00	0.98
BA-4Motifs	class 0	-	-	-	-	-	0.308	0.85	1.50	1.00	1.00	-	-	-	-	-	0.321	0.83	1.48	1.00	1.00
	class 1	-	-	-	-	-	0.235	0.83	1.38	1.00	0.66	-	-	-	-	-	0.235	0.83	1.38	1.00	0.66
	class 2	-	-	-	-	-	0.939	0.86	1.73	0.74	1.00	-	-	-	-	-	0.955	0.86	1.73	0.75	1.00
	class 3	-	-	-	-	-	0.743	0.85	3.25	0.21	0.91	-	-	-	-	-	0.743	0.85	3.23	0.21	0.91
	Any	0.713	0.85	1.60	0.99	0.94	0.777	0.85	1.66	0.99	0.87	0.723	0.85	1.53	1.00	0.94	0.777	0.85	1.63	1.00	0.87
BBBP	class 0	-	-	-	-	-	0.797	0.58	2.90	1.00	0.99	-	-	-	-	-	-	-	-	-	-
	class 1	-	-	-	-	-	0.743	0.54	3.08	1.00	0.96	-	-	-	-	-	-	-	-	-	-
	Any	0.810	0.57	3.00	1.00	0.98	0.786	0.57	2.93	1.00	0.99	-	-	-	-	-	-	-	-	-	-
Twitter	class 0	-	-	-	-	-	0.188	0.19	4.54	0.98	0.94	-	-	-	-	-	0.161	0.17	4.13	0.99	0.91
	class 1	-	-	-	-	-	0.469	0.22	4.04	0.99	0.97	-	-	-	-	-	0.384	0.18	3.23	0.99	0.95
	class 2	-	-	-	-	-	0.131	0.15	4.98	0.98	0.90	-	-	-	-	-	0.123	0.15	4.43	0.97	0.89
	Any	0.423	0.20	4.47	0.99	0.96	0.425	0.20	4.38	0.99	0.95	0.360	0.17	3.70	1.00	0.91	0.362	0.17	3.71	1.00	0.92
Graph-SST5	class 0	-	-	-	-	-	0.071	0.10	4.36	0.98	0.94	-	-	-	-	-	0.059	0.09	3.94	0.97	0.88
	class 1	-	-	-	-	-	0.205	0.16	4.18	0.94	0.93	-	-	-	-	-	0.181	0.17	3.72	0.95	0.92
	class 2	-	-	-	-	-	0.188	0.18	4.04	0.96	0.95	-	-	-	-	-	0.166	0.17	4.06	0.96	0.93
	class 3	-	-	-	-	-	0.234	0.16	3.69	0.97	0.96	-	-	-	-	-	0.213	0.16	3.53	0.96	0.94
	class 4	-	-	-	-	-	0.067	0.11	3.97	0.98	0.82	-	-	-	-	-	0.064	0.10	4.15	0.97	0.86
	Any	0.479	0.15	3.58	0.99	0.94	0.478	0.16	3.67	0.99	0.94	0.426	0.15	3.51	0.98	0.92	0.425	0.15	3.49	0.99	0.92

Denoising The cutoff is determined as a small fraction of the area under the curve of the sorted probabilities in with $p = 0.05\%$ for synthetic and molecular datasets, $p = 3\%$ for Twitter, and $p = 5\%$ for Graph-SST5. We selected these values so that the denoising step retains approximately $10 - 20\%$ of the original edges in the resulting graph, G_{DeN}.

Robustness For the BBBP dataset, we only consider edge removals when injecting noise. This is because adding edges would require introducing additional edge attributes.

A.6 Additional experiments

The per-class results when the OH module is used for Validity, Fidelity, Explanation Size, Minimality, and Motif Proximity are presented in Table 5. The module facilitates the model to learn associations between characteristic patterns in graphs and their corresponding classes, which is particularly useful for counterfactual search in multiclass settings. Consequently, at inference time, when conditioned on the desired counterfactual class, it can guide the addition of edges to form a new class-specific pattern that is sufficient to alter the prediction.



(a) cycle motif → house motif (b) cycle motif → clique motif (c) cycle motif → clique motif (d) cycle motif → clique motif

Fig. 1: Counterfactual examples generated by the model with the +OH module, illustrating the edge additions required to transform the original cycle motif into the OH target class.

As an illustrative example, consider an input graph G from the BA-3Motifs dataset that is initially classified as class 0, meaning that it contains a cycle over nodes $20 - 24$. Generating a counterfactual with predicted counterfactual class 1 requires transforming this cycle into a house motif, which can be achieved with a single edge addition. Accordingly, we observe that when conditioning the reconstruction model with the +OH module on class 1, the counterfactual explanations for graphs starting from class 0, typically consist of one or two added edges. One such counterfactual for G is shown in Fig. 1a, where the added edge that induces the prediction change is highlighted in green. By contrast, producing a counterfactual with predicted class 2 requires transforming this cycle into a clique motif, which requires five edge additions. Consistently, when conditioning the model on class 2, more counterfactuals are produced under the same threshold, since more candidate edges exceed it. Moreover, we observe counterfactuals with 4 or 5 added edges, **which does not occur when conditioning on class 1** (Figs. 1b–1d). Without the +OH module, the predicted edges still tend to appear around the motif region, which explains the relatively strong quantitative metrics, but they are noticeably less class-specific.

A.7 The Mutag dataset

In our experiments, we also included the MUTAG dataset following the Mutag0 variant used in [5]. The Mutag0 is extracted by treating benzene-NO_2 as the ground-truth explanatory motif for mutagenicity: mutagenic molecules contain benzene-NO_2, while non-mutagenic molecules do not. Class 0 corresponds to the mutagenic label and class 1 to the non-mutagenic label. The node features are encoded as one-hot vectors of the node labels. In our setup, we set the upper bound of deletions to $R = 1$ and disabled edge additions by setting $K = 0$, allowing the removal of at most one edge per graph. This proved sufficient to obtain counterfactuals for all graphs in the mutagenic → non-mutagenic direction, as deleting even a single important edge was enough to flip the prediction.

We did not consider edge additions (either alone or in combination with edge deletions), since reconstructing the ground-truth motif-and thus moving from the non-mutagenic to the mutagenic class-would generally require adding nodes (and not only edges) to explicitly create the motif. This highlights the need to incorporate domain knowledge, ensuring that the proposed edits remain chemically meaningful (i.e., they construct valid motifs) rather than flipping the prediction through arbitrary perturbations that merely degrade the graph structure.

In the mutagenic $\rightarrow$ non-mutagenic direction, our method achieves a validity of 1, an explanation size of 1, a fidelity score of 0.78, and a minimality score of 1, using the configuration parameters presented in Table 4. Tables 1 and 2 present the dataset statistics and the training hyperparameters for the oracle.

A.8 Experimental Details for Competing Baseline Methods

This section reports the training configurations and dataset-specific hyperparameters used for all baseline methods included in our evaluation. Unless otherwise reported, we follow the original implementations and recommended settings of each method. It is important to note that, although some explainers reported results on some datasets we also explore, we re-trained all baselines because our train/validation/test splits differ from those used in the original evaluations.

Random Baseline For each graph, we explore all combinations of $r = 0, 1, \ldots, R$ edge deletions and $k = 0, 1, \ldots, K$ edge additions until a counterfactual example is found. This approach is computationally expensive and does not scale to larger graphs while often producing CfXs that are not plausible; nevertheless, it provides a useful reference point against which the performance of approximate methods can be contrasted.

GIST For all datasets, GIST is trained for 50 epochs using a batch size of 16. The model uses hidden dimension 16 with 2 attention heads and the regularization parameter $\alpha = 0.9$. Optimization is performed with Adam using a learning rate of 10^{-3} and weight decay of 10^{-5}.

RSGG-CE For all datasets, RSGG-CE is trained for 30 epochs using the a positive and negative edge sampler with 500 sampling iterations. The framework trains one GAN per class. The generator and discriminator are optimized using SGD with learning rate 10^{-3} and batch size 4. Training uses binary cross-entropy loss. Dataset-specific maximum node count and embedding dimension are reported in Table 6.

CF^2 To support multiclass settings, counterfactual constraints are defined using class-relative margins with respect to the currently predicted class. Probability terms are temperature-scaled ($T = 5$) to prevent saturation in highly confident predictions. Across datasets, we use $\alpha = 0.7$, $\gamma = 0.9$, threshold 0.5, and 500 optimization epochs. The explainer mask is optimized with Adam using dataset-specific learning rate lr and sparsity weight λ (Table 7).

Table 6: Dataset-specific hyperparameters for RSGG-CE.

Dataset	Max nodes	Embedding dim
BA-2Motifs	25	4
BA-2Motifs-3Classes	25	4
BA-3Motifs	25	4
BA-4Motifs	25	4
BBBP	132	4
Graph-SST5	56	28
Twitter	73	28

Table 7: Dataset-specific hyperparameters for CF^2.

Dataset	λ	lr
BA-2Motifs	20	0.02
BA-2Motifs-3Classes	100	0.20
BA-3Motifs	100	0.02
BA-4Motifs	100	0.02
BBBP	20	0.05
Graph-SST5	20	0.05
Twitter	20	0.05

GCFExplainer GCFExplainer is a random-walk search baseline (not gradient training), therefore no optimizer, learning rate, or training epochs are used for the explainer itself. Table 8 reports the values we use for the decision threshold θ, teleport probability, neighborhood sampling size, and the maximum number of random-walk steps. We use $\alpha = 0.5$ for the coverage trade-off.

D4Explainer The explainer is trained for 800 epochs using Adam (learning rate 10^{-3}, betas $(0.9, 0.999)$, weight decay 0) together with an exponential learning-rate scheduler with decay $\gamma = 0.999$. The optimization objective is

$$\mathcal{L} = \mathcal{L}_{dist} + \alpha_{cf}\mathcal{L}_{cf},$$

with $\alpha_{cf} = 0.5$ and sparsity weight 2.5 in the BCE term.

The diffusion architecture uses 6 layers, 1 layer per convolution block, hidden size 64, instance normalization, concatenated outputs enabled, residual connections disabled, dropout 0.001, and noise MLP enabled. For counterfactual generation, we sample 150 candidates per graph.

A.9 System specifications

All experiments were run on a single workstation with an Intel Core i9-14900KS CPU (24 physical cores, 32 threads), 62 GB system RAM, and one NVIDIA GeForce RTX 4090 GPU.

Table 8: Dataset-specific hyperparameters for GCFExplainer. Maximum steps are target class-dependent; bracketed values correspond to each class within the dataset.

Dataset	θ	Teleport	Sample size	Max steps
BA-2Motifs	0.05	0.20	40	[4300, 5700]
BA-2Motifs-3classes	0.05	0.20	40	[5000, 3300, 5000]
BA-3Motifs	0.05	0.20	40	[3600, 3000, 3400]
BA-4Motifs	0.05	0.20	40	[2100, 2700, 2800, 2400]
BBBP	0.05	0.15	30	[14700, 46800]
Graph-SST5	0.15	0.60	20	[50000, 50000, 50000, 50000, 50000]
twitter	0.15	0.60	20	[50000, 50000, 50000]

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